

MANSFIELD LAHM, OH, USA

WMO: 724200

Lat: 40.820N

Lon: 82.518W

Elev: 393

StdP: 96.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14891

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-14.5	-21.0	0.6	-16.6	-18.3	0.8	-13.8	12.2	3.2	11.4	2.7	4.0	260	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	31.2	22.8	30.0	22.1	28.7	21.4	24.3	29.2	23.5	27.9	22.7	26.8	4.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.4	26.8	22.1	17.6	26.0	21.3	16.7	25.1	75.3	29.4	71.7	28.1	68.6	26.7	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.7	9.2	8.3	-20.6	33.0	3.5	1.4	-23.2	34.0	-25.2	34.8	-27.2	35.6	-29.8	36.5
			-20.5	25.5	3.0	0.9	-22.7	26.2	-24.5	26.7	-26.2	27.3	-28.4	27.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-3.4	-1.9	3.0	9.7	15.6	20.4	22.2	21.5	18.0	11.6	5.2	-0.2	(6)
	DBStd	10.42	6.99	6.47	6.34	5.40	4.84	3.71	2.97	2.92	4.03	5.05	5.48	5.86	(7)
	HDD10.0	1590	418	336	233	70	8	0	0	0	1	41	164	320	(8)
	HDD18.3	3364	674	566	476	262	114	22	4	7	54	216	396	576	(9)
	CDD10.0	1666	2	3	17	62	181	313	379	356	241	92	18	3	(10)
	CDD18.3	398	0	0	1	3	29	85	124	104	44	8	0	0	(11)
	CDH23.3	3087	0	0	5	33	240	661	996	757	346	50	1	0	(12)
	CDH26.7	831	0	0	1	2	48	186	297	197	93	7	0	0	(13)
(14) Wind	WSAvg	4.0	5.0	4.7	4.6	4.7	4.0	3.5	3.1	2.9	3.2	4.0	4.5	4.6	(14)
(15) Precipitation	PrecAvg	1011	58	51	82	95	110	101	104	101	83	60	88	74	(15)
	PrecMax	1707	141	120	179	167	224	254	336	193	197	144	326	284	(16)
	PrecMin	554	10	7	30	19	38	14	24	16	18	11	18	19	(17)
	PrecStd	229	29	27	38	40	45	56	61	46	45	31	59	41	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	18.2	22.9	26.7	29.8	32.2	33.0	32.1	31.6	27.5	21.0	16.3	(19)
		MCWB	13.4	13.9	15.2	17.1	20.6	22.7	24.4	23.5	21.9	19.8	14.5	13.7	(20)
	2%	DB	13.1	13.6	19.0	23.9	28.0	30.4	31.2	30.5	29.4	24.5	17.9	13.6	(21)
		MCWB	11.7	10.9	13.4	15.6	19.6	22.0	23.2	22.8	21.2	17.7	13.7	11.4	(22)
	5%	DB	10.0	10.5	16.2	21.6	26.2	28.9	29.7	28.9	27.2	22.3	16.0	11.0	(23)
		MCWB	8.4	7.9	11.8	14.6	18.7	21.3	22.5	22.0	20.2	16.7	11.9	9.1	(24)
	10%	DB	6.5	7.3	13.2	19.0	24.0	27.3	28.3	27.4	25.1	19.9	13.7	8.0	(25)
		MCWB	4.6	5.1	9.5	13.0	17.8	20.4	21.6	21.1	18.8	15.6	10.4	6.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	14.6	16.5	19.1	22.6	24.4	25.8	25.3	24.0	21.7	16.9	14.5	(27)
		MCDB	15.0	17.6	21.2	24.0	28.0	29.9	31.2	30.0	29.8	25.7	19.5	16.0	(28)
	2%	WB	11.8	11.4	14.3	17.0	21.1	23.1	24.6	24.1	22.5	19.6	14.5	11.9	(29)
		MCDB	13.1	13.0	18.0	22.0	26.0	28.3	29.4	28.5	26.9	22.7	17.0	12.9	(30)
	5%	WB	8.7	8.0	12.5	15.5	19.9	22.4	23.6	23.1	21.4	17.8	12.7	9.2	(31)
		MCDB	9.7	10.1	15.6	19.7	24.5	27.3	27.9	26.8	25.1	21.1	15.2	10.7	(32)
	10%	WB	4.7	5.1	10.2	14.0	18.7	21.5	22.7	22.2	20.2	16.0	10.7	6.3	(33)
		MCDB	6.4	7.2	13.2	18.1	23.0	25.8	26.7	25.7	23.7	19.3	13.4	8.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.0	9.2	11.1	10.8	10.4	10.4	10.2	10.8	10.0	8.2	6.9	(35)
		MCDBR	9.6	11.5	12.9	13.9	12.8	11.9	11.4	11.3	12.5	12.6	10.9	9.8	(36)
	5% WB	MCWBR	8.8	9.6	8.4	8.0	6.4	5.4	5.0	5.0	5.7	6.8	7.6	8.5	(37)
		MCWBR	9.4	11.0	11.5	12.0	10.9	10.3	9.9	9.6	10.2	10.8	9.8	9.8	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.299	0.325	0.348	0.386	0.419	0.438	0.443	0.426	0.378	0.355	0.320	0.302	(40)
		taud	2.420	2.332	2.354	2.270	2.213	2.199	2.203	2.235	2.351	2.403	2.467	2.445	(41)
	Ebn at Noon	Ebn at Noon	860	883	902	885	861	842	833	837	858	838	828	826	(42)
		Edh at Noon	80	101	111	130	141	143	141	133	111	94	77	73	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.49	2.34	3.40	4.61	5.45	5.98	5.96	5.37	4.33	2.89	1.88	1.26	(44)
	RadStd	RadStd	0.14	0.31	0.32	0.40	0.46	0.43	0.35	0.24	0.38	0.24	0.27	0.15	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+114	+62

Nomenclature: See separate page